

LinkedIn AI Auto-Poster

A self-hosted n8n workflow that researches, writes, illustrates and publishes LinkedIn posts with Gemini agents, after email approval

Complete Project Documentation

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Platform	n8n (self-hosted, Docker / VPS)
AI model	Google Gemini (all agents)
Image generation	Cloudflare Workers AI (FLUX)
Storage	Google Sheets
Publishing target	Personal LinkedIn profile
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Contents

1. Project Overview
2. Architecture
3. Prerequisites and Accounts
4. Google Sheet Schema
5. Workflow Node-by-Node
6. Agent Prompts
7. Image Generation (Provider History and Final Setup)
8. Approval Flow
9. Setup Guide
10. Testing Checklist
11. Troubleshooting
12. Security Best Practices
13. Maintenance and Future Improvements

1. Project Overview

The LinkedIn AI Auto-Poster is an automation built on n8n that produces and publishes LinkedIn posts with very little manual effort. Three Gemini-powered agents work in sequence: one chooses a topic and target audience, one writes the post, and one creates a matching image. Before anything goes live, the finished post and image are emailed for approval. Every published post is logged in a Google Sheet so topics are never repeated.

Goals

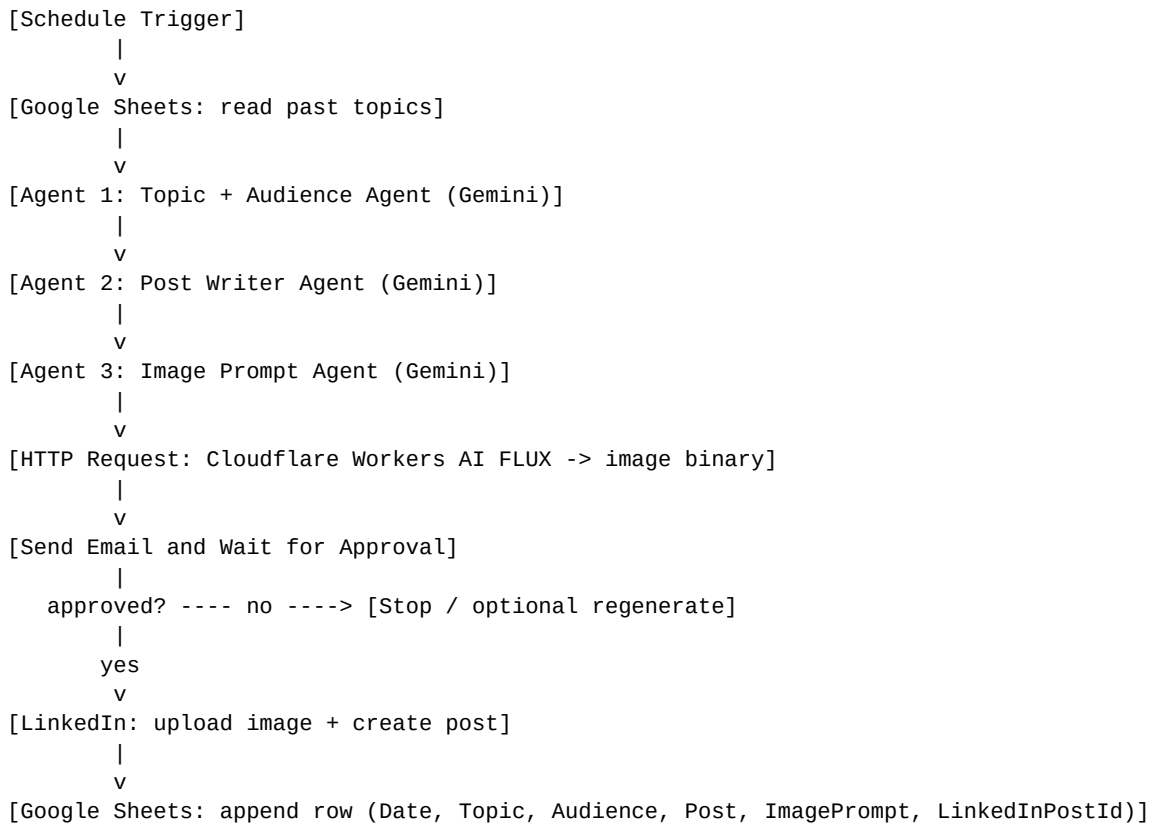
- Post consistently on LinkedIn without writing every post by hand.
- Make posts reflect the author's own point of view, goals and interests (personal context is supplied to every agent).
- Keep a natural, human, friends-conversation tone that does not sound AI-generated.
- Use images that match the post content and look natural rather than artificial.
- Keep a human in the loop: nothing is published without email approval.
- Avoid duplicate topics by checking and logging against a Google Sheet.
- Run on free or low-cost tiers wherever possible.

Key Design Decisions

Decision	Choice	Reason
Orchestrator	n8n, self-hosted	Full control, no execution limits, data stays on own server
LLM	Gemini for all agents	Single provider, generous free tier, consistent behaviour
Human oversight	Email approval before posting	Prevents off-brand or incorrect posts from going live
Duplicate protection	Google Sheet log	Simple, visible, easy to edit by hand
Target	Personal LinkedIn profile	Personal-brand building
Images	Cloudflare Workers AI FLUX	Free, good quality; earlier options hit limits (see section 7)

2. Architecture

The workflow is a linear pipeline with one approval gate. Data flows from a schedule trigger through three AI agents, to an email approval step, and finally to LinkedIn and the Google Sheet.



Components

Component	Role
n8n	Runs the workflow, schedules it, holds credentials
Gemini (Google AI Studio API)	Powers the topic, writer and image-prompt agents
Google Sheets	Memory of previously posted topics and a full post log
Cloudflare Workers AI	Generates the image from the image prompt (FLUX model)
Email (SMTP or Gmail node)	Sends the draft for approval and waits for the reply
LinkedIn API	Publishes the approved post with its image

3. Prerequisites and Accounts

Requirement	Details
n8n instance	Self-hosted via Docker on a VPS or local server, reachable over HTTPS (needed for OAuth callbacks and the approval link)
Google AI Studio API key	For Gemini. Create at aistudio.google.com
Google account	OAuth2 credential in n8n for Google Sheets (and optionally Gmail)
LinkedIn developer app	Needs the Share on LinkedIn and Sign In with LinkedIn products; OAuth2 credential in n8n
Cloudflare account	Account ID plus a Workers AI API token (see section 7 - an R2 token will not work)
Email sender	SMTP credentials or Gmail OAuth2 for the approval email
Personal context file	A text block with the author's goals, interests, opinions and writing style, pasted into each agent's system prompt

Recommended Docker Compose for n8n

```
services:
  n8n:
    image: docker.n8n.io/n8nio/n8n
    restart: always
    ports:
      - "5678:5678"
    environment:
      - N8N_HOST=n8n.yourdomain.com
      - N8N_PROTOCOL=https
      - WEBHOOK_URL=https://n8n.yourdomain.com/
      - GENERIC_TIMEZONE=Asia/Kolkata
      - TZ=Asia/Kolkata
    volumes:
      - n8n_data:/home/node/.n8n
volumes:
  n8n_data:
```

Put a reverse proxy (Caddy or Nginx) with a TLS certificate in front of n8n. LinkedIn OAuth and the approval email link both require a public HTTPS address.

4. Google Sheet Schema

Create one sheet (for example named **Posts**) with the header row below. The first row must match exactly what the n8n Google Sheets nodes map to.

Column	Content	Notes
Date	Date and time of posting	Use ISO format, e.g. 2026-10-01
Topic	Topic chosen by Agent 1	Read by Agent 1 next time to avoid repeats
Auidience	Target audience	The header is spelled this way in the live sheet. Either keep it consistently in every node, or rename the header to Audience and update the node mapping.
Post	Final post text	Exactly what was published
ImagePrompt	Prompt sent to the image model	Useful for reproducing or improving images
LinkedInPostId	ID returned by LinkedIn	Lets you find or delete the post later

How duplicates are avoided

- A Google Sheets node reads the Topic column (for example the last 50 rows) at the start of every run.
- That list is passed to Agent 1 with the instruction to choose a topic that is clearly different from all of them.
- Only after the post is successfully published is a new row appended, so rejected drafts do not pollute the log.

5. Workflow Node-by-Node

#	Node	Type	Purpose
1	Schedule Trigger	Trigger	Starts the run (for example Mon, Wed, Fri at 9:00 AM IST). Manual trigger is also kept for testing.
2	Get Past Topics	Google Sheets - Get Rows	Reads previous topics from the sheet.
3	Aggregate Topics	Aggregate / Code	Turns the rows into one list string for the prompt.
4	Topic Agent	AI Agent + Gemini Chat Model	Returns JSON: topic, audience, angle.
5	Post Writer Agent	AI Agent + Gemini Chat Model	Writes the post in a human tone using the topic, audience and personal context.
6	Image Prompt Agent	AI Agent + Gemini Chat Model	Creates a detailed, realistic image prompt that matches the post.
7	Generate Image	HTTP Request	Calls Cloudflare Workers AI FLUX and returns an image.
8	Convert Image	Convert to File / Code	Converts the base64 response to binary data for attaching and uploading.
9	Approval Email	Send Email and Wait for Response	Sends post text and image; waits for Approve / Reject.
10	IF Approved	IF	Continues only when approved.
11	Publish to LinkedIn	LinkedIn node or HTTP Request	Uploads the image and creates the post on the personal profile.
12	Log Post	Google Sheets - Append Row	Writes Date, Topic, Audience, Post, ImagePrompt, LinkedInPostId.

Structured output

Agents 1 and 3 should return strict JSON so downstream nodes can reference fields reliably. Use the n8n Structured Output Parser, or ask for JSON only in the prompt and parse it in a Code node.

```
// Agent 1 output
{ "topic": "...", "audience": "...", "angle": "..." }
```

```
// Agent 3 output
{ "image_prompt": "..." }
```

6. Agent Prompts

These are working templates. Replace the bracketed block with your own personal context (goals, interests, opinions, background). The same personal context goes into all three agents so the whole pipeline stays consistent.

6.1 Topic and Audience Agent

SYSTEM:

You are the content strategist for [AUTHOR NAME]'s personal LinkedIn.
Here is the author's background, goals and interests:
<<< PERSONAL CONTEXT >>>

Pick ONE topic for the next post that:

- fits the author's goals and interests above
- is useful or interesting to a clear audience
- is clearly different from these previously posted topics:
{{ past_topics }}

Return ONLY JSON:

```
{"topic": "", "audience": "", "angle": ""}
```

6.2 Post Writer Agent

SYSTEM:

You write LinkedIn posts as [AUTHOR NAME], in first person.
Personal context and point of view:
<<< PERSONAL CONTEXT >>>

Write like you are telling a friend something you genuinely learned.

Rules:

- Short and conversational. Simple words, short sentences.
- Open with a specific hook, not a generic line.
- Share a real opinion or lesson, not a list of tips.
- No corporate language, no buzzwords, no "In today's fast-paced world".
- No em dashes, no "Let's dive in", no "game-changer", no emoji spam (0-2 emojis at most).
- End with one natural question or thought, not a sales pitch.
- 80-150 words. 3-5 relevant hashtags at the very end.

USER:

Topic: {{ topic }}

Audience: {{ audience }}

Angle: {{ angle }}

6.3 Image Prompt Agent

SYSTEM:

You create image prompts for a text-to-image model (FLUX).
The image must match the meaning of the LinkedIn post and look like a real, natural photograph or clean editorial image, never obviously AI-generated.

Rules:

- Describe one clear subject and scene, setting, lighting, camera style (for example "35mm photo, natural window light, shallow depth of field").
- Natural colours and textures. No text, logos, watermarks or distorted hands/faces. Avoid neon, glossy 3D, and cliché robot or glowing-brain imagery.
- Keep it under 80 words.

Return ONLY JSON: {"image_prompt": ""}

USER:
Post: {{ post }}

7. Image Generation

Image generation was the part of the project that took the most iteration. The table records each provider tried, what happened, and the decision taken. It is kept here so the same dead ends are not repeated.

Order	Provider	Result	Outcome
1	Gemini image API	Free tier returned quota limit 0 (HTTP 429)	Abandoned for images; Gemini still used for text agents
2	Pollinations (URL-based, free)	HTTP 402 payment required	Abandoned
3	Pexels stock photos via search query	API key was paused, so no access	Abandoned for now
4	Cloudflare Workers AI - FLUX	First call returned 401 (auth error)	Current choice; fix is a correct API token (below)

Final setup: Cloudflare Workers AI

- In the Cloudflare dashboard, create an **API token** with the **Workers AI** permission (the 'Workers AI' template works). An R2 token causes the 401 error seen earlier because it lacks this permission.
- Note your **Account ID** (shown in the dashboard sidebar).
- In n8n, create a Header Auth credential: name **Authorization**, value **Bearer YOUR_TOKEN**.

HTTP Request node settings

Method: POST

URL: `https://api.cloudflare.com/client/v4/accounts/<ACCOUNT_ID>/ai/run/@cf/black-forest-labs/flux-1-schnel`

Authentication: Header Auth (Authorization: Bearer <TOKEN>)

Body (JSON): `{ "prompt": "{ { $json.image_prompt } }", "steps": 6 }`

Response format: JSON

The response contains the image as a base64 string at `result.image`. Convert it to binary in a Code or Convert to File node before attaching it to the email and uploading it to LinkedIn.

```
// Code node (Run Once for Each Item)
const b64 = $json.result.image;
return {
  json: { image_prompt: $('Image Prompt Agent').item.json.image_prompt },
  binary: { data: await this.helpers.prepareBinaryData(
    Buffer.from(b64, 'base64'), 'post-image.jpg', 'image/jpeg' ) }
};
```

Tip: the free Workers AI allowance is limited per day. If a run fails with a rate or quota error, add a Retry On Fail setting on the HTTP node, or post without an image as a fallback branch.

8. Approval Flow

Every draft is emailed before publishing. This is the main safeguard against low-quality or incorrect posts.

What the email contains

- Topic, audience and the full post text.
- The generated image attached (or embedded).
- Two actions: **Approve** and **Reject**.

Behaviour

- Use n8n's **Send and Wait for Response** operation (Gmail or Send Email node) with approval type set to approve / disapprove.
- The workflow pauses until a button is clicked. Set a timeout (for example 24 hours); on timeout the run ends and nothing is posted.
- Approved: the workflow continues to LinkedIn. Rejected: the run stops and nothing is logged, so the topic can be chosen again later.
- The approver address is set in the email node. Use the same inbox that you check daily.
- The n8n instance must be reachable over HTTPS, otherwise the approval buttons will not work.

Optional improvement: allow the reviewer to edit the text. Use the custom form response type so the approver can adjust the post, and publish the edited version.

9. Setup Guide

Step 1: Prepare the server

Install Docker, deploy n8n with the compose file in section 3, and place it behind HTTPS.

Step 2: Create credentials

Add credentials in n8n for Gemini (API key), Google Sheets (OAuth2), LinkedIn (OAuth2), Cloudflare (Header Auth) and email.

Step 3: Create the Google Sheet

Add the header row from section 4 and note the document and sheet names.

Step 4: Add personal context

Write a clear block about goals, interests, opinions and writing style and paste it into the three agent prompts.

Step 5: Build the workflow

Add the nodes in the order of section 5, with the Gemini chat model attached to each AI Agent.

Step 6: Configure image generation

Follow section 7, including the Workers AI token and the base64-to-binary conversion.

Step 7: Configure approval

Add the Send and Wait email node and the IF node after it.

Step 8: Connect LinkedIn publishing

Post as the authenticated person with the image. Save the returned post ID.

Step 9: Test manually

Run with the manual trigger and walk through the checklist in section 10.

Step 10: Activate

Switch the workflow to Active so the schedule trigger runs.

10. Testing Checklist

Check	Expected result
Sheet read	Past topics are returned (or empty list on first run)
Topic agent	Valid JSON; topic not in the past list
Writer agent	Natural tone, within word count, hashtags at the end
Image prompt	Under 80 words, realistic, matches the post
Image node	HTTP 200; binary image present; image looks natural
Approval email	Arrives with text and image; buttons work
Reject path	Nothing posted, nothing logged
Approve path	Post appears on LinkedIn with image
Log	New row appended with all six columns including LinkedInPostId
Second run	A different topic is chosen

11. Troubleshooting

Symptom	Likely cause	Fix
Cloudflare 401 Unauthorized	Wrong token type (for example an R2 token) or missing Bearer prefix	Create a token with Workers AI permission; use Authorization: Bearer TOKEN
Cloudflare 429 / quota	Daily free allowance used	Wait for reset, retry later, or add a no-image fallback
Gemini image 429 (limit 0)	Image generation not available on the free tier	Do not use Gemini for images; use Workers AI
Pollinations 402	Service now requires payment	Do not use it
Pexels key paused	Key disabled by provider	Re-activate the key or continue with Workers AI
Agent returns text around JSON	Model added commentary	Use Structured Output Parser or strip with a Code node
Posts sound robotic	Prompt too generic	Strengthen the banned-phrases list and add 2-3 real example posts of your own
Repeated topics	Past topics not passed in	Check the Aggregate node output and that the Topic column is read
Approval link does not work	n8n not on public HTTPS or WEBHOOK_URL wrong	Fix reverse proxy and WEBHOOK_URL
LinkedIn 401 / token expired	OAuth token expired	Reconnect the LinkedIn credential
Image missing in LinkedIn post	Binary field name mismatch	Ensure the upload node reads the field named data
Sheet columns blank	Header spelling mismatch (Audience)	Match the header text exactly in the node mapping

12. Security Best Practices

- Never paste API keys or tokens into prompts, notes, screenshots or chats. Store them only in n8n credentials.
- If a key or token has ever been shared in plain text, rotate it and create a new one.
- Set N8N_ENCRYPTION_KEY and keep a backup of it; credentials cannot be decrypted without it.
- Enable authentication on n8n and keep it updated; do not expose the editor without a login.
- Use HTTPS only, and restrict server SSH access to key-based login.
- Give the Cloudflare token only the Workers AI permission.
- Keep the approval step enabled so no content is published unseen.
- Back up the n8n data volume and export the workflow JSON regularly.
- Respect LinkedIn's terms and API limits; do not increase posting frequency aggressively.

13. Maintenance and Future Improvements

Routine maintenance

- Weekly: skim the Sheet to confirm posts and IDs are logging correctly.
- Monthly: refresh the personal context block so topics follow current goals; review which posts performed well and tune the prompts.
- As needed: update n8n, rotate credentials, re-authorise LinkedIn when the token expires.

Ideas for later

- Add a Reject-with-feedback path that regenerates the post using the reviewer's comments.
- Pull LinkedIn engagement numbers into the Sheet and feed top-performing topics back to Agent 1.
- Add a fallback image provider branch so a quota error never blocks a post.
- Add a weekly content-calendar mode that plans several topics at once.
- Send an error notification (Telegram or email) from an n8n Error Trigger workflow.
- Add few-shot examples of the author's own best posts to improve voice matching.

End of document